

## Portfolio CSS Changes Guide (Beginner-Friendly)

This guide explains every layout and styling update made to your portfolio so you can learn the resp

### Quick Highlights (Read This First)

- [IMPORTANT] `main { max-width: 1200px; margin: 0 auto; }` is the core centering rule for desktop.
- [IMPORTANT] `height: auto` on mobile cards prevents cramped content and uneven sections.
- [IMPORTANT] `flex-wrap: wrap` lets rows stack naturally when screen width gets smaller.
- [IMPORTANT] Use classes (like `.header-button`) for repeated elements, not duplicate IDs.
- [IMPORTANT] Keep `prefers-reduced-motion` so fade animations remain accessibility-friendly.
- [IMPORTANT] The projects list now uses JS to show exactly 5 cards before scrolling.
- [TWEAK FAST] To tighten layout quickly, edit gap values first before changing structure.

### What Was Fixed

1. Desktop spacing felt uneven with too much open space on the right side.
2. Header and body layout looked uneven on phones.
3. There was no entry transition when opening the website.
4. The projects list could grow too long and push the page too much.

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#### 1) Centering the layout on large screens

##### Problem

Your sections were anchored to the left using `justify-content: flex-start`, and some cards had manual

##### Fix

- Added a centered container for all page content:
  - `main { max-width: 1200px; margin: 0 auto; }`
- Updated section rows (`.main-first`, `.main-second`, `.main-third`) to:
  - `justify-content: center`
  - `flex-wrap: wrap`
  - a cleaner, consistent gap
- Removed manual left-push margins on main cards.

##### Why it works

`max-width` prevents super-wide stretching, and `margin: 0 auto` centers the whole content block. Center

##### Keep in mind

- [IMPORTANT] If desktop looks too wide again, check `max-width` first.
- [IMPORTANT] If cards hug one side, check `justify-content` and any manual margins.

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#### 2) Making the header and body responsive for phones

##### Header improvements

##### Problem

- Header spacing relied on hard-coded margins.
- Buttons used duplicate IDs (id="Resume" repeated), which is invalid HTML.

#### Fix

- Replaced duplicate button IDs with a reusable class: .header-button.
- Renamed buttons to distinct labels:
  - Resume
  - Projects
  - Contact
- Converted header groups into flexible rows that wrap naturally.
- Added tablet and mobile rules so title, buttons, and icons align evenly.
- Added a follow-up header spacing pass to prevent large vertical gaps on tablet/mobile by:
  - reducing row gap in .header-title
  - stopping header groups from flex-growing (flex: 0 0 auto)
  - keeping mobile header groups at width: auto so they stay compact

#### Why it works

Classes are reusable and valid for multiple elements. Flex wrapping and mobile breakpoints let compo  
The compact spacing pass prevents controls from being pushed far apart when the header stacks.

#### Keep in mind

- [IMPORTANT] Duplicate IDs can cause unexpected styling and JS behavior.
- [IMPORTANT] On stacked layouts, avoid flex-grow unless you intentionally want large spacing.

#### Body/card improvements

##### Problem

- Several cards used fixed heights and strict minimum widths.
- On smaller devices this can create cramped content, uneven spacing, and overflow.

##### Fix

- Standardized card sizing with responsive flex values.
- On mobile (max-width: 768px), switched major cards to height: auto so content defines the height.
- Relaxed strict min-width behavior at smaller breakpoints.
- Adjusted text/image sizing and spacing for smaller screens.
- Improved stacking for intro section and list sections.

#### Why it works

Responsive layouts should prioritize fluid sizing. height: auto on mobile avoids clipping, and small

#### Keep in mind

- [IMPORTANT] Fixed heights are okay for desktop sections, but usually risky on mobile.
- [IMPORTANT] Strict min-width values can break small-screen layouts.

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### 3) Adding a simple fade-in on page load

##### Problem

The page appeared instantly without transition.

##### Fix

- Added a keyframe animation to the body:
  - starts at opacity: 0
  - ends at opacity: 1
  - duration: 0.7s
- Added reduced-motion support:
  - @media (prefers-reduced-motion: reduce) disables animation and transitions.

#### Why it works

Fade-in adds a gentle first impression. Reduced-motion support keeps the experience accessible for u

#### Keep in mind

- [IMPORTANT] Always pair animation with reduced-motion support.
- [TIP] Keep first-load transitions short (0.5s to 0.8s) so the site still feels snappy.

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### 4) Limiting project cards to 5 visible items with scroll

#### Problem

Project cards use different text lengths and image heights, so a CSS-only fixed height cannot always

#### Fix

- Added a JS file: Scripts/JS/projects-scroll.js.
- The script measures the first 5 .Projects-list cards and sets max-height on .Projects-border-list.
- Kept scrolling enabled with overflow-y: auto in Scripts/CSS/main.css.
- Loaded the script in index.html before </body>.

#### Why it works

Because card heights are measured at runtime, the list always shows exactly 5 full cards first. If t

#### Keep in mind

- [IMPORTANT] Change MAX\_VISIBLE\_PROJECTS in Scripts/JS/projects-scroll.js if you want a different v
- [IMPORTANT] The script recalculates on page load and resize, so it stays accurate across screen si

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### File-by-file changes

#### index.html

- Updated header buttons from duplicate IDs to class-based buttons.
- Changed labels to Resume, Projects, and Contact.
- Added <script src="Scripts/JS/projects-scroll.js"></script> so project list height is computed aut

#### Scripts/CSS/header.css

- Added page fade-in animation and keyframes.
- Centered header container with max width.
- Refactored button styling from #Resume to .header-button.
- Improved responsive behavior for tablet and mobile.
- Tightened stacked header spacing to remove excessive vertical space.
- Added reduced-motion fallback.

#### Scripts/CSS/main.css

- Added centered main container with max width.
- Centered row layouts and standardized spacing.
- Removed manual left offsets from bordered cards.
- Improved card width behavior for large screens.
- Added responsive breakpoints for tablet/phone.
- Switched card heights to auto on mobile.
- Enabled vertical scrolling in .Projects-border-list (overflow-y: auto).

Scripts/JS/projects-scroll.js

- Added runtime height calculation for .Projects-border-list.
- Shows exactly 5 project cards first, then scrolls for additional cards.
- Recalculates on DOMContentLoaded, load, and resize.

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## Quick CSS Concepts You Just Used

- [IMPORTANT] max-width + margin: 0 auto = centered content block on large screens.
- [IMPORTANT] display: flex + flex-wrap: wrap = responsive row that can stack naturally.
- [IMPORTANT] flex: 1 1 320px = grow and shrink, with a preferred base width.
- [IMPORTANT] Media queries (@media) = apply rules only for certain viewport widths.
- [IMPORTANT] height: auto = let content control the box height.
- @keyframes + animation = create entry transitions.
- prefers-reduced-motion = accessibility-friendly motion handling.
- JS measurement + max-height = exact visible card count with variable content.

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## How to tweak this later (safe edits)

If you want to customize the look further:

1. Wider/narrower desktop content:
  - edit main { max-width: 1200px; } in Scripts/CSS/main.css.
2. More/less spacing between cards:
  - edit gap in .main-first, .main-second, .main-third.
3. Faster/slower fade-in:
  - edit animation: pageFadeIn 0.7s ... in Scripts/CSS/header.css.
4. Earlier/later phone layout switch:
  - adjust values in media queries (900px, 768px, 480px).
5. Tighter/looser stacked header spacing:
  - adjust .header-title { gap: ... } and the tablet/mobile flex rules in Scripts/CSS/header.css.
6. More/less project cards visible before scrolling:
  - edit MAX\_VISIBLE\_PROJECTS in Scripts/JS/projects-scroll.js.

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## Final outcome

You now have:

- A centered, balanced desktop layout.
- A cleaner and more even phone/tablet layout.

- A simple and accessible page-entry fade-in.
- A projects list that shows 5 cards first, then scrolls for more.

These are core responsive CSS patterns used in real-world production websites.